

```
$ mysql -e "SHOW CREATE TABLE orders" --normalize
```



// 1NF → 2NF → 3NF □□□□□

1NF

□□□

2NF

□□□□□□

3NF

□□□□□□□

□□□□

OLAP □□□

\$ cat schema-plan.md



01 / 
□□□□□□□□

02 / 
1NF    → 2NF → 3NF

03 / 
JOIN □□□

04 / 
OLTP vs OLAP □□□

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PART 01



XXXXXXXX N

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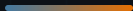
order_id	customer_name	customer_email	product_name	price
1	□□	<u>zhang@example.com</u>	□□□□□	30000
2	□□	<u>zhang@example.com</u>	□□	500
3	□□	<u>li@example.com</u>	□□	1200

□□□□

□□□□□□□□ email □□□□

□□□□

□□□ email□□ UPDATE □□□□



ROOT CAUSE



CHECKPOINT_01

データベースの UPDATE 37 回実行後 3 分経過

A データベースの更新が完了した

B データベースの更新が完了した

C データベースの更新が完了した

PART 02



1NF → 2NF → 3NF →

1NF

NO DUPLICATES

BEFORE —

student	courses
	Maths, Maths, Maths, Maths, Maths, Maths, Maths, Maths

AFTER — 1NF

student	course
	Maths
	Maths
	Maths

2NF NO GLYPH

BEFORE — □□□□

order_id	product_id	product_name	qty
1	P1	□□	1
2	P1	□□	3

AFTER — □□□□□

order_id	product_id	qty
1	P1	1
product_id	product_name	
P1	□□	

3NF

NO SLIP

□□□□□□

student_id → dept_id → dept_name | dept_name | dept_id | — |

BEFORE — □□□□

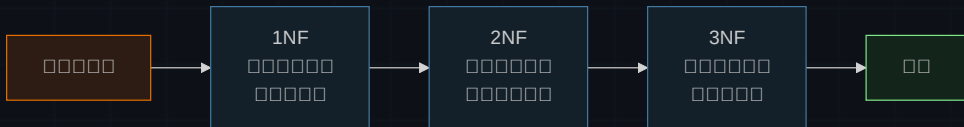
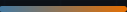
student_id	name	dept_id	dept_name
S1	□□	D1	□□□
S2	□□	D1	□□□

AFTER — □□□□□

student_id	name	dept_id
S1	□□	D1
dept_id	dept_name	
D1	□□□	

□□□□□□□□□□ — □□□□□□□□□□

□ □ □ □ □ □ □



3 columns 3NF 2 columns 6 columns 4 columns

PART 03



□□□□□□JOIN □□□

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JOIN □□
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JOIN □□□□□

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OLTP vs OLAP

OLTP	OLAP	Comparison
Transactional	Analytical	
Real-time	Batch	
High frequency	Low frequency	
Small transactions	Large transactions	
Simple queries	Complex queries	
OLTP	OLAP	

STRATEGY

OLTP is designed for high transaction volume and low latency. OLAP is designed for high data volume and high latency.



→



1NF     → 2NF     → 3NF    

3

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OLTP □□□□OLAP □□□□□□□□

Schema normalized.

XXXXXXXXXXXX JOIN XXXXX

```
$ mysql -e "EXPLAIN SELECT ..."
```

1NF → 2NF → 3NF → XXXXXXXX